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ORIGINAL ARTICLE

The Development of a Scoring System for the Kinetic House-Tree-Person Drawing Test

Chih-Ying Li [a], Tsyr-Jang Chen [b], Christine Helfrich [a], Ay-Woan Pan [c,d,*]

[a] *Sargent College of Health and Rehabilitation Sciences, Department of Occupational Therapy, Boston University, Boston, Massachusetts, USA*
[b] *Department of Mechanical Engineering of Lung Hwa University of Science and Technology, Tao-Yuan, Taiwan*
[c] *School of Occupational Therapy, College of Medicine, National Taiwan University, Taipei, Taiwan*
[d] *Department of Psychiatry, National Taiwan University Hospital, Taipei, Taiwan*

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validity

Abstract *Objective/Background:* The Kinetic House-Tree-Person Test (KHTPT) is already in widespread use amongst psychiatric occupational therapists in Taiwan, but the psychometric attributes of the test are somewhat limited. The primary aims of this study are to develop a quantitative scoring system for the KHTPT and carry out an assessment of its psychometric attributes.
Methods: Based on the test manual and the related literature, we identified 35 drawing characteristics relating to anxiety and depression, and we recruited 323 participants from two universities in northern Taiwan to participate in our study. These participants, who had a mean age of 20.1 years (SD = 3.0), were instructed to draw a KHTPT picture, and were then asked to complete related questionnaires. The data were subsequently analysed using WINSTEPS (Beavertown, Oregon: Winsteps.com) and SPSS (SPSS Inc., Chicago).
Results: The results revealed that all of the items of the scoring system provided a good fit with the Rasch measurement model, with the Cronbach's alpha for the scale being .94. The Spearman correlation coefficients of the Rasch-transformed KHTPT scores with the Beck Anxiety Inventory, the Beck Depression Inventory-II, the Beck Hopelessness Scale, and the National Taiwan University Hospital Symptom Checklist were all found to be small, albeit with statistical significance (Spearman correlation coefficients, $r = .140-.226$).
Conclusion: This study demonstrates our proposed KHTPT scoring system has eminently acceptable construct validity and internal consistency. We suggest that future studies should include patients with psychiatric disorders at varying degrees of severity or functional level to examine the applicability and predictive validity of our proposed scoring system.

* Corresponding author. National Taiwan University and the Department of Psychiatry, National Taiwan University Hospital, Room 407, Number 17, Xu-Zhou Road, Taipei, 10020, Taiwan.
E-mail address: aywoan@ntu.edu.tw (A.-W. Pan).

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Introduction

Occupational therapists use a variety of activities to interact with their patients (Crepeau, Cohn, & Schell, 2003; Hopkins, Smith, & Tiffany, 1978; Kielhofner, 1992; Slavson, 1934). Of the various therapeutic activities, art activities are particularly appropriate for creating a context within which patients can vent their emotions and appropriately express themselves (Creek, 1996; Dollin, 1976; Friedman, 1952, 1953; Frye, 1990; Giles, 1985; Gillette, 1963; Harries, 1992; Lloyd & Papas, 1999; Martin, 1991; Storr, 1972; Willson, 1983). Indeed, art activities have proven to be extremely therapeutic and have led to significant changes among various patients (Fukunishi et al., 2002; Guillemin, 2004; Palmer et al., 2000; Timothy, 1993; Zalsman et al., 2000; Zoltan, 1998).

In an attempt to capture the effects of art activities on the wellbeing of the patients, as well as the meanings behind the drawings, scholars have developed different ways of measuring the characteristics of their patients through their drawings (Bonder 1993; Chie & Haruo, 2004; Groth-Marnat & Roberts, 1998; Lally, 2001; Polatajko & Kaiserman, 1986; Thompson and Blair, 1998).

Delatte and Hendrickson (1982) utilised human figure drawings (H-F-D) to detect the relationship between size of drawings and self-esteem of the junior high school students. The results showed that the correlation among height, superficial coverage, and self-esteem was significant for male students (Chie & Haruo, 2004).

Goodenough—Harris developed a Draw-A-Man Test (D-A-P) in 1926 in which scholars suggested that D-A-P could be used to estimate the intelligence of children (Harris, 1963). Research also found that certain drawing characteristics of the Kinetic Family Drawing (K-F-D) are related to psychological status. For example, unstable lines of the drawings may represent the use of decompensatory defense; shadows and overturn circles in the drawing may indicate psychotic symptoms. (Virshup, 1976).

The House-Tree-Person (H-T-P) drawing test, developed by Buck (1948) and modified by Hammer (1969), is a standardised instrument used to decompose the psychological state of the patients with mental illness. The H-T-P has been examined to determine its utility with an acute psychiatric inpatient (Polatajko & Kaiserman, 1986).

Among these, the Kinetic House-Tree-Person Test (KHTPT) has clearly become the most frequently used by occupational therapists throughout Taiwan (Hsiao, Pan, Chung, & Lu, 2000). It was also used by occupational therapists worldwide (Drake, Lo, Hwang, & Shih, 1994; Polatajko & Kaiserman, 1986). The KHTPT, which is used to evaluate the psychological state and self concept of an individual (Burns, 1987), is essentially a modification of the H-T-P drawing test, involving the incorporation of an interactive (kinetic) component that provides ample information of the state of the patients. However, there is a distinct lack of evidence on the psychometric attributes of the KHTPT; as a result, those therapists applying KHTPT in their clinical practices have been hampered by a lack of evidence proving its validity and usefulness.

The purpose of the present study is therefore to develop a scoring system for the KHTPT and to examine the reliability and validity of this proposed scoring system. The set of items representing the drawing characteristics relating to the mood status of individuals, as well as the psychiatric symptoms of mental illness, have already been collated and assessed by researchers based on the drawing test manual and the related literature (Burns, 1987; Groth-Marnat & Roberts, 1998; Marzolf & Kirchner, 1973; Palmer et al., 2000; Zalsman et al., 2000).

These drawing characteristics are included in our scoring system based on the description criteria relating to depression or anxiety. We ultimately identified a total of 35 items, for which a score of 1 was to be allocated if any of these characteristics was displayed. A higher KHTPT score would indicate a greater probability of emotional distress.

The researchers in the present study carried out two preliminary studies of the scoring system. The first study involved the recruitment of 137 college students, with 44 of these participants completing the KHTPT twice over a 2-week period for the purpose of test—retest reliability. The results of the first study revealed that the intraclass correlation coefficient (ICC) of the scoring system was .72, with one-half of the 35 items being found to have kappa values above .40.

In accordance with the unidimensionality requirement of Rasch analysis, all of the items were found to provide good fit, whilst the item difficulty logits were found to range between 13.7 and 76. The KHTPT scores had significant correlations with the total scores of the Beck Anxiety Inventory (BAI), the Beck Depression Inventory, Version 2 (BDI-II), and the National Taiwan University Hospital Symptoms Checklist (NTUHSC), which included a few items from the BAI and the BDI-II [$r = .31–.84$ (Pan, Chen, & Li, 2006)].

The second study involved the recruitment of 81 participants from a public college, with the results revealing a negative correlation between the KHTPT and the perceived level of satisfaction with the quality of life (QOL) amongst individual college students. Specifically, the total KHTPT score was found to have significant correlations with a few of the BDI-II items (loss of energy, tiredness, or fatigue) and also some of the BAI items (unable to relax, unsteady, and nervous) ($r = .216–.335$). The KHTPT score was also found to be capable of predicting 10.5% of mood status and the QOL score (Li, Pan, Chen, Chung, & Hsiung, 2006). Additional participants were recruited into the study to ensure that the KHTPT items provided a good fit with the Rasch measurement model as well as reliable calibration.

The primary aims of this study are to examine the reliability and validity of the KHTPT scoring system amongst a group of college students in Taiwan. Specifically, our research focuses on the unidimensionality of the KHTPT, internal consistency, and concurrent validity.

Methods

Participants and procedure

This study involved the recruitment of a total of 323 participants from two universities in northern Taiwan. All of the participants were instructed to draw a KHTP picture with a pencil on an approximately A 4 size (8.5″ × 11″) piece of white paper placed horizontally in front of them. The specific instruction given was, "Please draw a house, a tree, and a whole person on this piece of paper including some kind of action. Try to draw a whole person, not a cartoon or stick person." Afterward, the participants answered questions about their drawings and completed the BAI, the BDI-II, the Beck Hopelessness Scale (BHS) (Beck & Steer, 1988), the NTUHSC, and the World Health Organization Quality of Life Questionnaire—Brief Version (WHOQOL-BREF) (Yao, Chung, Yu, & Wang, 2002). This study follows the ethical principles of the Declaration of Helsinki for medical research involving humans.

Instruments

KHTPT scoring system

We identified a set of drawing characteristics relating to depression and anxiety through the test manual and the extant literature; this resulted in the identification of 35 items. These items were subsequently divided into the four domains of drawing characteristics: (1) general; (2) house; (3) tree; and (4) person (Appendix I). Each item was allocated a score of 1 if certain characteristics were found to be present; any test revealing a total score of 35 would indicate the most severe degree of depression/anxiety of the drawer. The scoring system was programmed into the computer using Visual Basic software (Microsoft Corporation, 1998).

BAI

The BAI is a 21-item self-reported rating scale intended to assess the severity of anxiety among adults and adolescents aged 13 years or older. This index was found to have high internal consistency with a Cronbach's α of .92—.94, and test—retest reliability of .75 (Beck & Steer, 1987).

BDI-II

The BDI-II is a 21-item self-reported rating scale used to assess the severity of depression in adults and adolescents aged beyond 13 years. It is one of the most frequently used indicators of depression severity among both researchers and clinicians. This index was found to have high internal consistency, with a Cronbach's α of .92—.93, and test-retest reliability of 0.93. The construct validity was evidenced by the factor analysis approach using the Rasch measurement model (Beck, Steer, & Brown, 1996; Pan & Hsu, 2008).

BHS

The BHS is a 20-item self-reported rating scale used to assess the severity of negative attitudes or pessimism among adults and adolescents; the scale is in widespread use for persons with mental illness who are also classified as being at risk for suicide. This scale was found to have high internal consistency, with a Cronbach's α of .82—.93, and test—retest reliability of 0.66—0.90. The concurrent validity was found to be 0.62—0.74, with the construct validity being evidenced through the adoption of the factor analysis approach (Beck & Steer, 1988).

NTUHSC

The NTUHSC is a 50-item self-reported rating scale designed to identify and assess the severity of psychiatric symptoms among patients with mental illness in Taiwan (Huang, 1998; Huang, Hung, & Shieh, 1999; Yang, Tseng, Chung, & Yip, 1992). The NTUHSC was originally developed based on the Symptom Checklist-90 (SCL-90) introduced by Derogatis et al. (1973). The checklist comprises of 50 items under the 10 subscales of somatisation, obsession, interpersonal sensitivity, depression, anxiety, hostility, phobia, paranoia, psychoticism, and other symptoms (Tsai et al. 1978).

WHOQOL-BREF

The WHOQOL-BREF is a 28-item self-report rating scale measuring the following four domains: physical health (seven items), psychological health (six items), social relationships (four items) and environment (nine items), and overall perception of quality of life and health (two items). The WHOQOL-BREF, which is a brief version of the original instrument, assesses the perceptions of individuals in the context of their culture and value systems, and has proven to be more convenient for use in large research studies or clinical trials. The WHOQOL-BREF is based on a five-point Likert-type scale, with the total respective scores available for each of the four domains being 35, 30, 20, and 45.

A reliability study revealed that the ranges for the four domains were Cronbach's alpha values of .74—.82, composite reliability of .79—.84, and test—retest reliability of .61—.79. Furthermore, a validity study revealed highly significant correlations between the items and their corresponding domains ($r > 0.4$, $p < 0.01$), which were found to be much larger than the correlations between the same items and other domains (Yao et al., 2002).

Data analysis

SPSS (Version 11.5) was used to analyse all of the data in this study, with Spearman correlation coefficients being applied to study the concurrent validity of the KHTPT. WINSTEPS (Version 3.56) was also adopted for our examination of the construct validity of the KHTPT, including both the validity and sensitivity of the items. The Rasch measurement model, which is based on single parameter item response theory was further applied in order to study the psychometric attributes of the KHTPT.

The Rasch measurement model computer programme (WINSTEPS) facilitates the transformation of ordinal data into interval data, such that the extent of certain latent traits among individuals can be effectively placed on a continuum relative to the location of the items; this process offers additional advantages, insofar as each item can be calibrated more accurately and used to create a tailored computerised adaptive test (Linacre, 2006).

The adequacy of the fit of each item is assessed by infit mean square (MNSQ) statistics, which provide information on whether the responses to the item measures conform to measurement theory (Lai et al., 2005). An infit MNSQ measure of less than 1.0 indicates dependency, whereas an infit MNSQ measure greater than 1.0 indicates noise. The acceptable values of infit MNSQ range from 0.6 to 1.4. Finally, the item and person separation index (G) was also used to examine how many strata the items and persons can be separated by the persons and items tested; the formula used in the present study was $(4\,G + 1)/3$ (Linacre, 2006).

Table 1 Demographics of the Participants ($N = 323$)

Variables	No. of obs.	Missing values	Mean	SD
Sex				
Male	113 (35%)	—	—	—
Female	210 (65%)	—	—	—
Age	301	22	20.07	3.02
Kinetic-House-Tree-Person Drawing Test	323	—	4.39	2.34
Beck Anxiety Inventory	323	—	7.33	6.44
Beck Depression Inventory-II	323	—	9.01	7.81
Beck Hopelessness Scale	236	87	4.08	3.38
National Taiwan University Hospital Symptom Checklist	120	203	84.18	28.52
WHO Quality of Life Questionnaire—Brief Version (WHOQOL-BREF)	80	243	76.49	15.37

— = NA.

Table 2 Item Calibration and Fit Statistics of the KHTPT Scoring System Items ($N = 323$)[a]

Item	Contents	Raw score	Measure	Infit statistics	
				MNSQ	ZSTD
16	Stained glass window	1	85.1	1.00	0.3
30	Clothes too big	8	64.0	1.00	0.0
3	Very short length; circling	1	76.6	1.01	0.0
27	Black coloured hands	3	65.4	0.99	0.0
31	Big shoes	3	65.4	0.98	0.0
15	Emphasis on the bedroom	3	65.4	1.00	0.0
11	Lines drawn at the top of the paper	4	62.4	1.02	0.0
29	Disproportionately small body	6	58.2	1.01	0.0
7	A "birds' eye" view	7	56.5	1.00	0.0
32	Repeatedly erased	7	56.5	1.03	0.1
34	Parts of the extremities coloured black	7	56.5	1.00	0.0
24	Big ears, darkened ears seen through transparent hair	9	53.8	1.02	0.1
33	Rotated figure	9	53.8	1.03	0.1
14	Blinds on windows	9	53.8	1.01	0.1
18	Dead root	9	53.8	0.96	−0.1
21	Swaying tree trunk	9	53.8	0.96	−0.1
8	Within the grids	10	52.7	1.06	0.2
17	Dead branch	10	52.7	0.96	−0.1
6	Located at the bottom of the paper	11	51.6	1.04	0.1
23	Very big eyes (filled in shadow)	13	49.8	1.08	0.4
5	Very small drawing size	17	46.7	1.05	0.3
35	Shadow painting	17	46.7	0.90	−0.5
22	Very small head	19	45.4	1.02	0.1
1	Unusually weak strength	20	44.7	0.99	0.0
19	Paint shadows on the trunk	21	44.2	1.04	0.2
9	Drawn on the border	23	43.0	1.06	0.4
13	Netted roof	29	40.1	0.96	−0.3
10	Object crosses the border	35	37.6	1.11	1.0
25	No mouth	35	37.6	0.96	−0.4
4	Shadow	38	36.4	0.85	−1.7
20	Very thin trunk	38	36.4	0.96	−0.4
2	Discontinuous and curved lines	43	34.6	0.92	−1.0
26	Blurred or lightened hands	44	34.2	0.90	−1.2
28	Thin feet	46	33.6	1.14	1.7
12	No chimney	119	11.3	0.99	−0.1

[a] The scores are listed in 'measure' order.

Results

As shown in Table 1, the mean age of the 323 participants was 20.1 years (SD = 3.0); the majority of the sample population, 210 (65%), were female. The average KHTPT score was found to be 4.4 (SD = 2.3), with a range of between 0 and 13, whilst the internal consistency of the KHTPT scoring system (Cronbach's alpha) was found to be .94.

Unidimensionality of the KHTPT

Table 2 reveals that, as expected, all 35 items provided a good fit with the Rasch measurement model, constituting a continuum of items capable of representing the different degrees of depression or anxiety. This result demonstrates that the items in the KHTPT form a unidimensional construct.

Frequency of endorsement

Table 2 also shows that a higher calibration for any particular item denotes a lower frequency of endorsement for that item. The five most rarely displayed drawing characteristics were "stained glass window," "clothes too big," "big shoes," "parts of the extremities coloured black," and "black-coloured hands." The five most frequently occurring drawing characteristics were "no chimney," "blurred or lightened hands," "thin feet," "discontinuous and curved lines," and "no mouth" (Table 2). Most of the items were distributed at the higher end of the continuum.

Concurrent validity

Although the Spearman correlation coefficients on the KHTPT, the BAI, the BDI-II, the BHS, and the NTUHSC were all found to be low, they were nevertheless significant, ranging between 0.140 and 0.226 (Table 3).

Item-total correlation

The item-total correlation ranged between 0.02 and 0.43, whilst the internal consistency of the items was 0.612.

Person and item separation index

The G was 0.67, indicating that we can differentiate the persons into at least one statistically distinct strata. The item separation index was 4.03, indicating that we can differentiate the items into at least six statistically distinct strata (Fig. 1).

Table 3 Spearman Correlation Coefficients on Concurrent Validity of the KHTPT Scoring System (N = 323)

Variables	KHTP test total score
Beck Anxiety Inventory	0.140*
Beck Depression Inventory-II	0.192**
Beck Hopelessness Scale	0.224**
National Taiwan University Hospital Symptom Checklist	0.226*
WHO Quality of Life Questionnaire—Brief Version (WHOQOL-BREF)	−0.160

* Indicates significance (p) at the .05 level.
** Indicates significance (p) at the .01 level (two-tailed test).

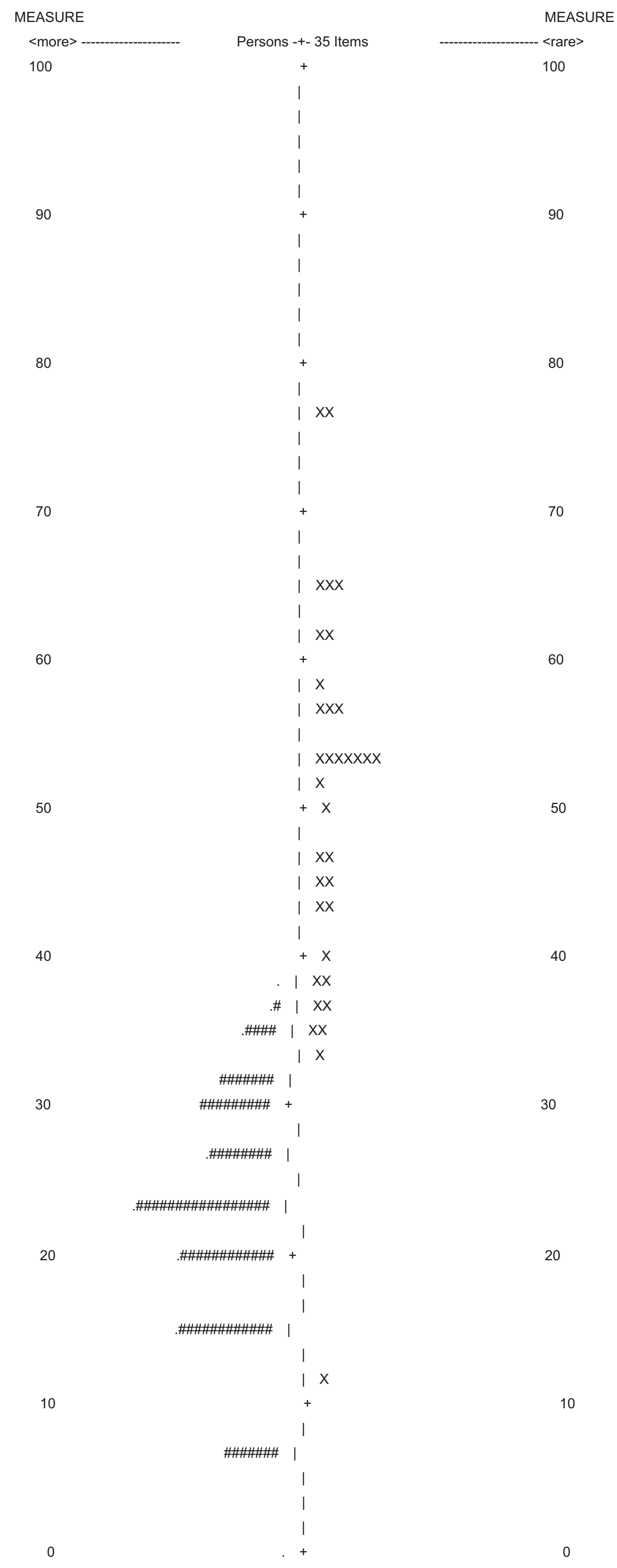

Figure 1 Map of 323 college students and 35 items.

Discussion

This study demonstrates that the internal consistency and construct validity of our proposed KHTPT scoring system are within acceptable parameters, thereby indicating that these items can be spread along a continuum ranging from those KHTPT drawing characteristics that are presented least often to those that are presented most often. As expected, the items within the scale are correlated to a moderate degree.

The results of this study further demonstrate that cultural differences may have some influence on those items that are found to appear more or less frequently; for example, the "no chimney" item was found to be the most prevalent characteristic amongst all of the drawings. This is particularly relevant because it fits in with the weather in Taiwan, which is warm and humid all year long, so there are very few households that feature a chimney. There were also very few drawings featuring stained glass windows, since this is more relevant to Western art forms and cultures.

The low but significant results of the KHTPT with regard to these measures may indicate that the KHTPT is unique in its ability to identify the characteristics of the drawers; this is quite distinct from self-reported symptom severity rating scales or the QOL scale (Palmer et al., 2000; Yang et al., 1992; Yao et al., 2002). Patients may use the KHTPT to vent their feelings toward certain objects, whereas it may not be representative of their mood status or level of anxiety.

Limitations

There are a few limitations of the present study that must be taken into consideration. First of all, the study participants were college students, so it may be safe to assume that the vast majority had no mental illness, which would, of course, reduce the reliability of the item calibration. Since the intended application of the KHTPT is to aid therapists' understanding of their patients' psychological state and self-concept in clinical practice (Bonder, 1993; Dollin, 1976; Eklund, 2000; Frye, 1990; Giles, 1985; Harries, 1992), there is a requirement to replicate the study among a group of people with mental illness, with particular focus on those who have difficulty expressing their feelings or assessing themselves using a self-reported questionnaire.

Secondly, the interpretation of the KHTPT relies on both the characteristics of the drawing and the perception of the drawers (Burns, 1987). Since this study does not link the scoring system with other qualitative information, it may well fail to produce a meaningful interpretation of the KHTPT.

Conclusion

This study demonstrates that our proposed KHTPT scoring system has eminently acceptable construct validity and internal consistency. We suggest that future studies should include patients with psychiatric disorders at varying degrees of severity or functional level to examine the applicability and predictive validity of our proposed scoring system.

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Appendix I KHTPT Drawing Characteristics of Depression and Anxiety

Domain	Item no.	Drawing characteristics
General	1	Unusually weak strength
	2	Discontinuous and curved lines
	3	Very short length; circling
	4	Shadow
	5	Very small drawing size
	6	Located at the bottom of the paper
	7	A "birds' eye" view
	8	Within the grids
	9	Drawn on the border
	10	Object crosses the border
	11	Lines drawn at the top of the paper
House	12	No chimney
	13	Netted roof
	14	Blinds on windows
	15	Emphasis on the bedroom
	16	Stained glass window
Tree	17	Dead branch
	18	Dead root
	19	Paint shadows on the trunk
	20	Very thin trunk
	21	Swaying tree trunk
Person	22	Very small head
	23	Very big eyes (filled in shadow)
	24	Big ears; darkened ears seen through transparent hair
	25	No mouth
	26	Blurred or lightened hands
	27	Black coloured hands
	28	Thin feet
	29	Disproportionally small body
	30	Clothes too big
	31	Big shoes
	32	Repeatedly erased
	33	Rotated figure
	34	Parts of the extremities coloured black
	35	Shadow painting